

Robust Two-Stage Radiotherapy Treatment Planning

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Handling Uncertain Data With Robust Optimization

- ① We would like to assume that good data leads to good decisions.
- ② However, as reported in [1], for some LPs even a 0.01% perturbation in the data may cause certain constraints to become highly infeasible.

Several Key Observations

- 1 At the time of decision making, data isn't known exactly.
- 2 We cannot ignore this uncertainty.
- 3 A methodology for handling this uncertainty is needed.

Enter Robust Optimization (For the LP case)

A generic LP can be characterized with the triplet (A, b, c) .

$$\min_{x \in \mathbb{R}^n} \{c^T x : Ax \leq b\}$$

In the robust optimization framework, an *uncertain-LP* is defined as a collection:

$$\min_{x \in \mathbb{R}^n} \{c^T x : Ax \leq b, \forall (A, b, c) \in \mathcal{U}\}$$

What does it mean to solve an uncertain-LP?

The Underlying Decision-Making Environment

- ① x represents "here-and-now" decisions.
- ② The decision-maker is fully responsible for the consequences of the decisions, only when the actual data is within $\mathcal{U}$.
- ③ The decision-maker cannot tolerate violations of the constraints when the data is in $\mathcal{U}$.

An "uncertainty immunized" solution

In this environment, an uncertainty immunized solution is a fixed vector that remains feasible whatever the realization of the data within $\mathcal{U}$.

We call such a solution *robust feasible*.

What is the value of a *robust feasible* solution?

Since we want a solution that is feasible no matter the realization, it is natural to attribute the worst-case value of the uncertain-LP to the robust feasible solution, that is:

$$\sup\{c^T x : (A, b, c) \in \mathcal{U}\}$$

Thus, the best *robust feasible* solution can be found by solving the following LP:

$$\min_x \left\{ \sup_{(A,b,c) \in \mathcal{U}} c^T x : Ax \leq b, \forall (A,b,c) \in \mathcal{U} \right\}$$

or equivalently:

$$\min_{x,t} \{t : c^T x \leq t, Ax \leq b, \forall (A, b, c) \in \mathcal{U}\}$$

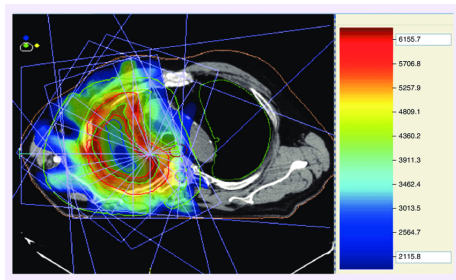
Discovery of X-rays

In 1895, Wilhelm Conrad Röntgen documented his discovery of X-rays, earning him the Nobel Prize in Physics in 1901.



Applications of X-rays

Today, Röntgen's X-rays are used in a variety of applications, including Radiotherapy.



Planning a Radiotherapy Treatment

- 1 We aim to deliver a high radiation dose to the tumor.
- 2 We would like to deliver a low radiation dose to healthy tissues.
- 3 We want the radiation dose delivered to the tumor to be relatively uniform.

Optimizing a Radiotherapy Treatment Plan - Notation

- ① $[m]$ - the set of all voxel indices.
- ② $T \subseteq [m]$ - the set of all tumor voxels.
- ③ $H_k \subseteq [m]$ - the set of all voxels in the k -th healthy organ, $\forall k \in [K]$.
- ④ n - the number of beamlets whose intensity is to be determined.
- ⑤ $\mathbf{D} \in \mathbb{R}^{m \times n}$ - the dose matrix, where $\mathbf{D}_{i,j}$ is the influence the j -th beamlet has on the i -th voxel.
- ⑥ N - the number of fractions.
- ⑦ $\mathbf{x} \in \mathbb{R}^{N \times n}$ - the intensity of the n beamlets in each of the N fractions.

Note: $T \cup \bigcup_{k=1}^K H_k = [m]$

What's Uncertain?

The so-called *physical dose* applied to a voxel v at fraction f is given by:

$$d_{f,v}(\mathbf{x}) = \sum_{i=1}^n \mathbf{D}_{v,i} x_{f,i}$$

where $\mathbf{D}_{v,i}$ is the influence the i -th beamlet has on the voxel v .

However, the impact of the *physical dose* also depends on biological factors, such as the oxygenation level of the voxel.

This information is modeled via a *radiosensitivity vector* $\Phi \in [0, 1]^{N \times T}$.

Accordingly, the *bio-adjusted dose* applied to a voxel v at fraction f is given by:

$$\phi_{f,v} \cdot d_{f,v}(\mathbf{x})$$

Unfortunately, Φ is estimated from medical images, that are taken days or weeks before the treatment, and so, we only have an uncertain estimate, $\hat{\Phi}$.

Optimizing a Radiotherapy Treatment Plan – Nominal Formulation

Taking $\hat{\Phi}$ at face value, we formulate the following LP, which optimizes the radiotherapy treatment plan in accordance with the aforementioned goals:

$$\max_{\mathbf{x} \in \mathbb{R}_+^{N \times n}, \underline{d} \in \mathbb{R}, \underline{d}^F \in \mathbb{R}} \quad \underline{d} + \lambda \underline{d}^F \quad (1.a)$$

$$\text{subject to} \quad \underline{d}^F \leq \hat{\phi}_{fv} d_{fv}(\mathbf{x}), \quad v \in T, f \in [N] \quad (1.b)$$

$$\hat{\phi}_{fv} d_{fv}(\mathbf{x}) \leq \mu^F \underline{d}^F, \quad v \in T, f \in [N] \quad (1.c)$$

$$\underline{d} \leq \sum_{f \in [N]} \hat{\phi}_{fv} d_{fv}(\mathbf{x}), \quad v \in T \quad (1.d)$$

$$\sum_{f \in [N]} \hat{\phi}_{fv} d_{fv}(\mathbf{x}) \leq \mu \underline{d}, \quad v \in T \quad (1.e)$$

$$d_{fv}(\mathbf{x}) \leq \bar{d}_k^F, \quad f \in [N], k \in [K], v \in H_k \quad (1.f)$$

$$\sum_{f \in [2]} d_{fv}(\mathbf{x}) \leq \bar{d}_k, \quad k \in [K], v \in H_k \quad (1.g)$$

Optimizing a Radiotherapy Treatment Plan – Uncertainty Set

Here we apply the robust optimization methodology by assuming that the values of Φ are members of the following uncertainty set, henceforth referred to as the *spatially and temporally bound uncertainty set*:

$$\mathcal{U}_{\text{STB}} = \left\{ \Phi \in [0, 1]^{N \times |T|} \left| \begin{array}{ll} |\phi_{1v} - \hat{\phi}_{1v}| \leq \delta, & \forall v \in T \\ |\phi_{fv} - \phi_{fu}| \leq \gamma_{vu}, & \forall f \in [N], v, u \in T \\ |\phi_{fv} - \phi_{(f+1)v}| \leq \sigma, & \forall f \in [N-1], v \in T \end{array} \right. \right\}$$

- δ - the maximum allowed deviation from the measured radiosensitivity vector.
- γ_{vu} - parameters given by a metric-preserving function of the distance between v and u . Limits the maximum allowed deviation between the radiosensitivity values of two consecutive fractions.
- σ - the maximum allowed deviation between the radiosensitivity value of a voxel across two consecutive fractions.

Optimizing a Radiotherapy Treatment Plan – Robust Formulation

After a couple of simple algebraic manipulations, we can write the *robust counterpart* of the nominal LP:

$$\max_{\mathbf{x} \in \mathbb{R}_+^{N \times n}, \underline{d} \in \mathbb{R}, \underline{d}^F \in \mathbb{R}} \underline{d} + \lambda \underline{d}^F \quad (2.a)$$

$$\text{subject to } \min_{\Phi \in \mathcal{U}_{\text{STB}}} \{ \phi_{fv} d_{fv}(\mathbf{x}) \} \geq \underline{d}^F, \quad v \in T, f \in [M] \quad (2.b)$$

$$\max_{\Phi \in \mathcal{U}_{\text{STB}}} \{ \phi_{fv} d_{fv}(\mathbf{x}) - \mu \phi_{fu} d_{fu}(\mathbf{x}) \} \leq 0, \quad v, u \in T, f \in [M] \quad (2.c)$$

$$\min_{\Phi \in \mathcal{U}_{\text{STB}}} \left\{ \sum_{f \in [M]} \phi_{fv} d_{fv}(\mathbf{x}) \right\} \geq \underline{d}, \quad v \in T \quad (2.d)$$

$$\max_{\Phi \in \mathcal{U}_{\text{STB}}} \left\{ \sum_{f \in [M]} \phi_{fv} d_{fv}(\mathbf{x}) - \mu \sum_{f \in [M]} \phi_{fu} d_{fu}(\mathbf{x}) \right\} \leq 0, \quad v, u \in T \quad (2.e)$$

$$d_{fv}(\mathbf{x}) \leq \bar{d}_k^F, \quad f \in [M], k \in [K], v \in H_k \quad (2.f)$$

$$\sum_{f \in [2]} d_{fv}(\mathbf{x}) \leq \bar{d}_k, \quad k \in [K], v \in H_k \quad (2.g)$$

Optimizing a Radiotherapy Treatment Plan – Reformulation

The robust counterpart is a semi-infinite program — that is, a program with an infinite number of constraints — and must be reformulated to be solvable.

Lemma 1

Let $\mathcal{U} \subseteq \mathbb{R}^n$ be a Reformulated Robust set. Then for any continuous function $f : \mathbb{R}^{|\mathcal{I}|} \rightarrow \mathbb{R}$,

$$\min_{\mathbf{u}_{\mathcal{I}} \in \mathcal{U}} \{f(\mathbf{u}_{\mathcal{I}})\} = \min_{\mathbf{u}_{\mathcal{I}} \in \mathcal{P}_{\mathcal{I}}(S)} \{f(\mathbf{u}_{\mathcal{I}})\}$$

Lemma 2

The projection of $\mathcal{U}_{STB}$ onto ϕ_{fv} is

$$P_{fv}(\mathcal{U}_{STB}) = \left[\min_{\phi_{fv} \in \mathcal{U}_{STB}} \{\phi_{fv}\}, \max_{\phi_{fv} \in \mathcal{U}_{STB}} \{\phi_{fv}\} \right] \quad \forall v \in T, f \in [2]$$

Optimizing a Radiotherapy Treatment Plan – 1-D Projections

Lemma 3

Consider the case where $N = 2$, and suppose for any $u, v \in T$, we have $\gamma_{uv} = \Gamma(\text{dist}(u, v))$, where $\text{dist}(u, v)$ is the voxel-to-voxel distance, and $\Gamma : \mathbb{R} \rightarrow [0, 1]$ is a nondecreasing, subadditive function satisfying $\Gamma(x) = 0 \iff x = 0$. Define for all $v \in T$:

$$\underline{\phi}_{1v}^0 = \max \{0, \hat{\phi}_{1v} - \delta\}, \quad \bar{\phi}_{1v}^0 = \min \{\hat{\phi}_{1v} + \delta, 1\} \quad (1)$$

$$\underline{\phi}_{1v} = \max_{u \in T} \{\underline{\phi}_{1u}^0 - \gamma_{uv}\}, \quad \bar{\phi}_{1v} = \min_{u \in T} \{\bar{\phi}_{1u}^0 + \gamma_{uv}\} \quad (2)$$

$$\underline{\phi}_{2v} = \max \{0, \underline{\phi}_{1v} - \sigma\}, \quad \bar{\phi}_{2v} = \min \{\bar{\phi}_{1v} + \sigma, 1\} \quad (3)$$

Then,

$$\max_{\phi \in \mathcal{U}_{STB}} \phi_{fv} = \bar{\phi}_{fv}, \quad \min_{\phi \in \mathcal{U}_{STB}} \phi_{fv} = \underline{\phi}_{fv}$$

Optimizing a Radiotherapy Treatment Plan - Reformulated Constraint 2.b

Corollary 1

Constraint 2.b can be rewritten as:

$$\underline{\phi}_{fv} \cdot d_{fv}(\mathbf{x}) \geq \underline{d}^F, \quad v \in T, f \in [2] \quad (3.b)$$

Optimizing a Radiotherapy Treatment Plan - Reformulated Constraint 2.d

Corollary 2

Constraint 2.d can be rewritten as:

$$\sum_{f \in [N]} \underline{\phi}_{fv} \cdot d_{fv}(\mathbf{x}) \geq \underline{d} \quad (3.d)$$

Optimizing a Radiotherapy Treatment Plan – 2-D first fraction

Lemma 4

Suppose $v \in T$, and let $\underline{\phi} \in \mathbb{R}^{2 \times |T|}$, $\bar{\phi} \in \mathbb{R}^{2 \times |T|}$ have components given by (2) and (3), and suppose that for any $u, v \in T$,

$$\gamma_{uv} = \Gamma(\text{dist}(u, v)),$$

where $\text{dist}(u, v)$ is the voxel-wise distance, and $\Gamma : \mathbb{R} \rightarrow [0, 1]$ is a nondecreasing, subadditive function satisfying $\Gamma(x) = 0 \iff x = 0$.

Then, the projection of $\mathcal{U}_{STB}$ onto (ϕ_{1v}, ϕ_{1u}) satisfies:

$$\mathcal{P}_{1v, 1u}(\mathcal{U}_{STB}) = \mathcal{U}_{v, u}^{(1)} \equiv \left\{ (\phi_{1v}, \phi_{1u}) \in \mathbb{R}^2 : \underline{\phi}_{1v} \leq \phi_{1v} \leq \bar{\phi}_{1v}, \underline{\phi}_{1u} \leq \phi_{1u} \leq \bar{\phi}_{1u}, |\phi_{1v} - \phi_{1u}| \leq \gamma_{vu} \right\}$$

Optimizing a Radiotherapy Treatment Plan – 2-D second fraction

Lemma 5

Suppose $v \in T$, and let $\underline{\phi} \in \mathbb{R}^{2 \times |T|}$, $\bar{\phi} \in \mathbb{R}^{2 \times |T|}$ have components given by (2) and (3). Suppose also that for any $u, v \in T$,

$$\gamma_{uv} = \Gamma(\text{dist}(u, v)),$$

where $\text{dist}(u, v)$ is the distance between voxels u and v , and $\Gamma : \mathbb{R} \rightarrow [0, 1]$ is a nondecreasing, subadditive function satisfying $\Gamma(x) = 0 \iff x = 0$.

Then, the projection of $\mathcal{U}_{STB}$ onto (ϕ_{2v}, ϕ_{2u}) is:

$$\mathcal{P}_{2v, 2u}(\mathcal{U}_{STB}) = \mathcal{U}_{v, u}^{(2)} \equiv \left\{ (\phi_{2v}, \phi_{2u}) \in \mathbb{R}^2 : \underline{\phi}_{2v} \leq \phi_{2v} \leq \bar{\phi}_{2v}, \underline{\phi}_{2u} \leq \phi_{2u} \leq \bar{\phi}_{2u}, |\phi_{2v} - \phi_{2u}| \leq \gamma_{vu} \right\}$$

Optimizing a Radiotherapy Treatment Plan - Reformulated Constraint 2.c

Observation 1

Invoking proposition 7 from [2], we can replace constraint 2.c by the constraint pair:

$$\bar{\phi}_{fv} \cdot d_{fv}(\mathbf{x}) + \max \left\{ \underline{\phi}_{fu}, \bar{\phi}_{fv} - \gamma_{uv} \right\} \cdot (-\mu^F d_{fu}(\mathbf{x})) \leq 0, \quad v, u \in T, f \in [N] \quad (2.c_1)$$

$$\max \left\{ \underline{\phi}_{fu} + \gamma_{uv}, \bar{\phi}_{fv} \right\} \cdot d_{fv}(\mathbf{x}) + \underline{\phi}_{fu} \cdot (-\mu^F d_{fu}(\mathbf{x})) \leq 0, \quad v, u \in T, f \in [N] \quad (2.c_2)$$

Reformulated Robust Optimization Problem (Without commulative homogeneity)

$$\max_{\mathbf{x} \in \mathbb{R}_+^{2 \times n}, \underline{d} \in \mathbb{R}, \underline{d}^F \in \mathbb{R}} \underline{d} + \lambda \underline{d}^F \quad (3.a)$$

$$\text{subject to } \underline{\phi}_{fv} \cdot d_{fv}(\mathbf{x}) \geq \underline{d}^F, \quad v \in T, f \in [2] \quad (3.b)$$

$$\bar{\phi}_{fv} \cdot d_{fv}(\mathbf{x}) + \max \left\{ \underline{\phi}_{fu}, \bar{\phi}_{fv} - \gamma_{uv} \right\} \cdot (-\mu^F d_{fu}(\mathbf{x})) \leq 0, \quad v, u \in T, f \in [2] \quad (3.c')$$

$$\max \left\{ \underline{\phi}_{fu} + \gamma_{uv}, \bar{\phi}_{fv} \right\} \cdot d_{fv}(\mathbf{x}) + \underline{\phi}_{fu} \cdot (-\mu^F d_{fu}(\mathbf{x})) \leq 0, \quad v, u \in T, f \in [2] \quad (3.c'')$$

$$\sum_{f \in [2]} \underline{\phi}_{fv} \cdot d_{fv}(\mathbf{x}) \geq \underline{d}, \quad v \in T \quad (3.d)$$

$$d_{fv}(\mathbf{x}) \leq \bar{d}_k^F, \quad f \in [2], k \in [K], v \in H_k \quad (3.e)$$

$$\sum_{f \in [2]} d_{fv}(\mathbf{x}) \leq \bar{d}_k, \quad k \in [K], v \in H_k \quad (3.f)$$

Reformulated Robust Optimization Problem

Numerical Experiments

Future Directions

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- [2] Aditya P. Apte et al. *Robust Radiotherapy Planning with Spatially and Temporally Based Uncertainty Sets*. NSF Proposal #2024098. Available upon request. 2024.